

SAHAKAR-SETU: MASTER TECHNICAL SYSTEM ARCHITECTURE & DATA FLOW

Ministry of Cooperation | National Council for Cooperative Training (NCCT) | 45 Integrated Subsystem Components

LAYER 1: CLIENT ACCESS & PHYSICAL EDGE DEVICES

PANEL A: MOBILE & WEB CLIENTS

Trainee Mobile PWA

Next.js 14, Servist

IndexedDB Offline

Voice & Chatbot UI

MediaRecorder API

22 Indic Languages

Institute ERP Portal

React, TanStack Table

FullCalendar Sched.

Hostel & Mess UI

Dynamic Bed Grid

Coupon Disbursal

PACS Bulk Sponsor

Society Nomination

CSV Bulk Engine

Cooperative Jobs

Recruiter Portal

1-Click Apply

Ministry Cockpit

Apache Superset

Live Pan-India BI

PANEL B: PHYSICAL HARDWARE KIOSK (CLASSROOM DOOR)

8" Tablet Enclosure

Tamper-Proof VESA

5000mAh Battery

HD Optical Sensor

Wide Dynamic Range

Low-Light Enhanced

INT8 Face ONNX

<80ms Match in RAM

MiniFASNet Liveness

Offline SQLite

50,000 Event Queue

Dynamic QR Backup

HTTPS / TLS 1.3: Voice Audio (WebM), Phone OTP, Profile Updates, Bulk CSV Rosters

HMAC-SHA256 Encrypted Sync: 128-d Face Vectors, KioskID, Timestamps (No Raw Photos)

LAYER 2: UNIFIED API GATEWAY & REVERSE PROXY

Nginx / Cloudflare Edge

TLS 1.3 Strict Cipher

WAF & Rate Limiting (Token Bucket)

JWT & RBAC Multi-Institute Filter

Internal Reverse Proxy Dispatch (/api/v1/auth, /erp, /lms, /jobs, /creds)

LAYER 3: CORE APPLICATION MODULAR MONOLITH (FastAPI High-Performance Async Backend)

ERP & Academic Workflow

- Camunda BPMN State Machine
- Multi-Tier Sponsoring Approvals
- Conflict-Free Timetable Scheduler
- PostgreSQL pg_trgm Fuzzy Dedup

Hostel & Logistics Engine

- Dynamic Room & Bed Matrix Grid
- Gender & Date Batch Constraints
- Real-Time Check-In / Occupancy
- Mess Meal Coupon QR Issuance

Attendance & Stipend Engine

- Biometric Log Ingestion & Verification
- Automated TA/DA Stipend Calculator
- Anomaly & Duplicate Claim Gate
- Direct Benefit Transfer (DBT) Output

Multilingual LMS & Sandbox

- Audio-First Micro-Lessons (<2MB)
- 22-Module PACS ERP Simulator
- Client-Side Quiz Scoring Engine
- Synchronizes Integer Score Only

Cryptographic Credentials

- W3C Verifiable Credential Model
- Ed25519 64-Byte Digital Signature
- High-Density 2D QR Code Stamping
- Zero Blockchain / \$0 Gas Fees

Job Matching & Guidance

- Deterministic Candidate Ranker
- Score = w1(Grade)+w2(Dist)+w3(Rec)
- AI Voice Career Counseling Bot
- 1-Click Verified Transcript Apply

SQL with Row-Level Security (RLS) Tenant Isolation

Pub/Sub Event Bus & Celery Async Queues

Direct Presigned S3 URLs (Audio Chunks & Signed PDFs)

LAYER 4: PERSISTENCE & DATA STORAGE LAYER

Primary Relational DB: PostgreSQL 16

Multi-Tenant Row-Level Security (RLS) | pg_trgm Bi-Gram Matching | ACID Attendance Ledger

In-Memory Cache & Broker: Redis 7

Session Store | Celery Task Queues (Stipends, SMS, PDF Rendering) | Token Rate Limit Counters

Object Storage: Cloudflare R2 / AWS S3

Micro-Lesson Audio Bundles (<2MB) | Tamper-Proof Cryptographic Certificate PDFs with 2D QR

gRPC Audio Streams (ASR & TTS)

JSON-LD W3C Verifiable Credential

REST: Bidirectional Vacancy Sync

REST: PACS Registration Verification

REST: DLT Notification Payload

LAYER 5: NATIONAL DIGITAL PUBLIC INFRASTRUCTURE (DPI) & EXTERNAL SERVICES

Bhashini / AI4Bharat Gateway

- IndicWhisper Speech-to-Text (ASR)
- IndicTrans2 Neural Translation (NMT)
- Meta MMS-TTS Audio Generation (22 Lang)

DigiLocker & NAD API

- Permanent W3C Credential Vault
- Direct Push to Trainee Aadhaar Locker
- National Academic Depository Compliance

National Career Service (NCS)

- Ministry of Labour & Employment Portal
- Ingestion of 79,630 PACS Vacancies
- Auto-Sync of Certified Alumni Pool

National Cooperative DB (NCD)

- Master Registry of all Indian Cooperatives
- Verification of Genuine Sponsoring PACS
- Validates Legal Registration Numbers

NIC SMS & WhatsApp DLT Gateway

- Mandatory TRAI DLT Template Dispatch
- Transactional OTP & Admission Passes
- TA/DA Stipend Bank Transfer Notifications

■ EDGE ATTENDANCE: <80ms Face Match | 100% Offline SQLite | Zero Raw Photos (DPDP Act)

■ VERIFICATION: Ed25519 2D QR | Sub-15ms Offline Validation | \$0 Blockchain Gas Fees

■ INFRASTRUCTURE: Modular Monolith | Scales to 20 Institutes & 79,630 PACS on <\$90/mo VPS